Functions: Composite, Inverse & Logarithmic

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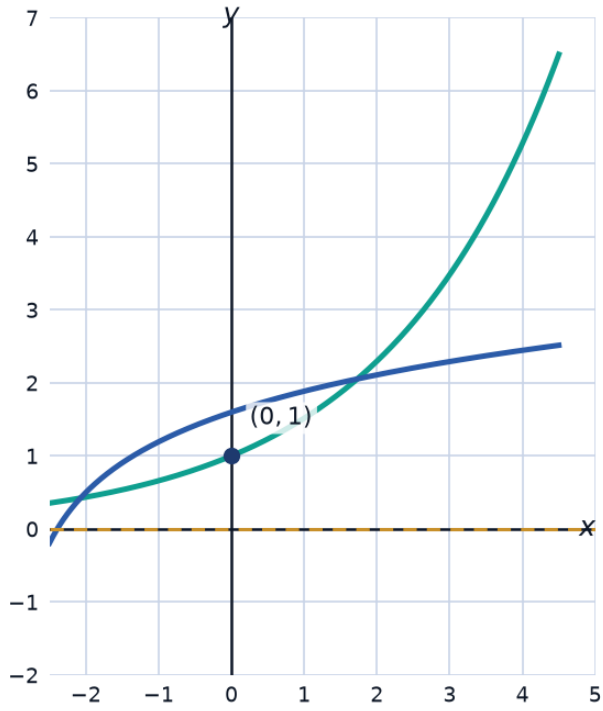
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Domain and range

Find the domain of $f(x) = \frac{1}{x^2 - 9}$.

2 [Foundational] • Target: Domain and range

Find the domain of $g(x) = \sqrt{7 - 2x}$.



3 [Foundational] • Target: Domain and range

Find the range of $f(x) = -(x - 2)^2 + 5$.

4 [Foundational] • Target: Composite functions

Given $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, find $(f \circ g)(x)$.

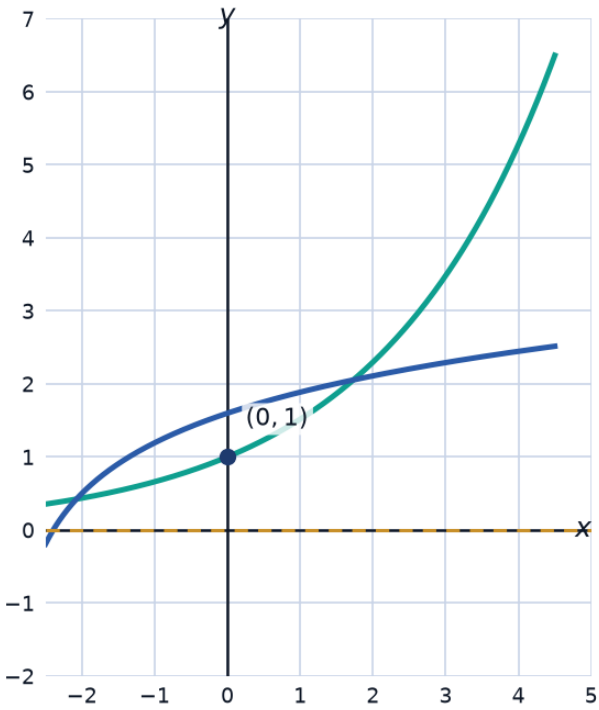
5 [Foundational] • Target: Composite functions

Given $f(x) = \sqrt{x + 4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(x)$.

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6 [Foundational] • Target: Composite functions

Using the same f and g as the previous question, find $(g \circ f)(x)$ and state its domain.



7 [Foundational] • Target: Inverse functions

Find $f^{-1}(x)$ for $f(x) = \frac{2x-5}{3}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Inverse functions

Find $f^{-1}(x)$ for $f(x) = x^3 + 2$.

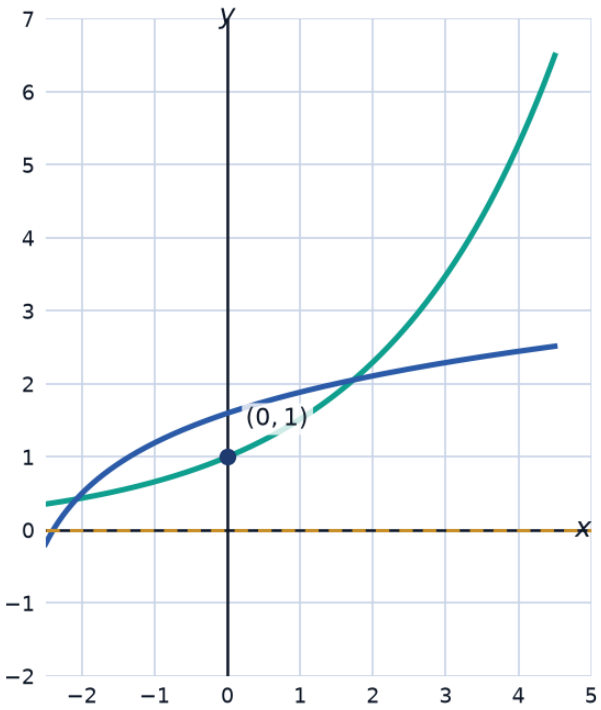
9 [Intermediate] • Target: Verifying inverses

Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x-1}{5}$ are inverses by showing $f(g(x)) = x$.

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10 [Intermediate] • Target: Verifying inverses

Does $f(x) = x^2$ (for all real x) have an inverse function? Explain.



11 [Intermediate] • Target: Logarithm laws

Expand $\log_b\left(\frac{x^3y}{z^2}\right)$ using the laws of logarithms.

12 [Intermediate] • Target: Logarithm laws

Condense $2\log_5 x + 3\log_5 y - \log_5 z$ into a single logarithm.

13 [Intermediate] • Target: Logarithm laws

Evaluate $\log_2 32 - \log_2 4$ without a calculator.

14 [Intermediate] • Target: Solving logarithmic equations

Solve $\log_4 (x^2) = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

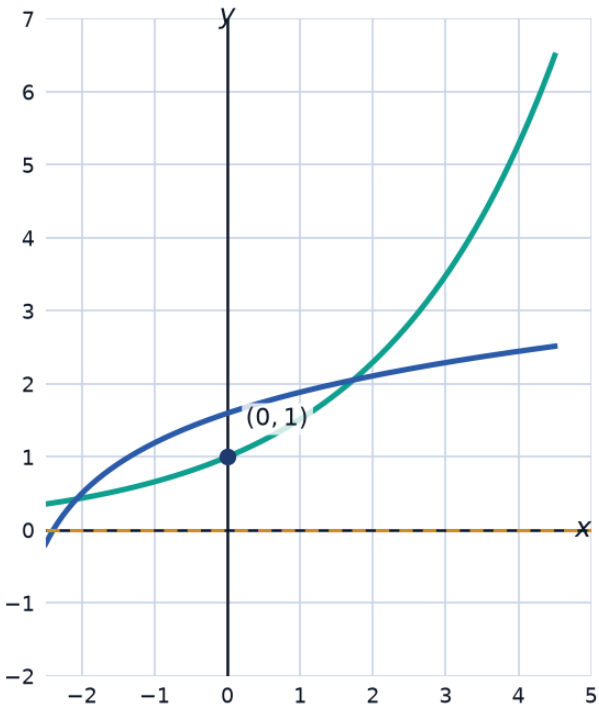
15 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Solve $\ln x + \ln(x - 3) = \ln 10$.

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16 [Challenging] • Target: Exponential equations and logs as inverses • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Solve $e^{2x-1} = 20$, giving your answer to three decimal places.



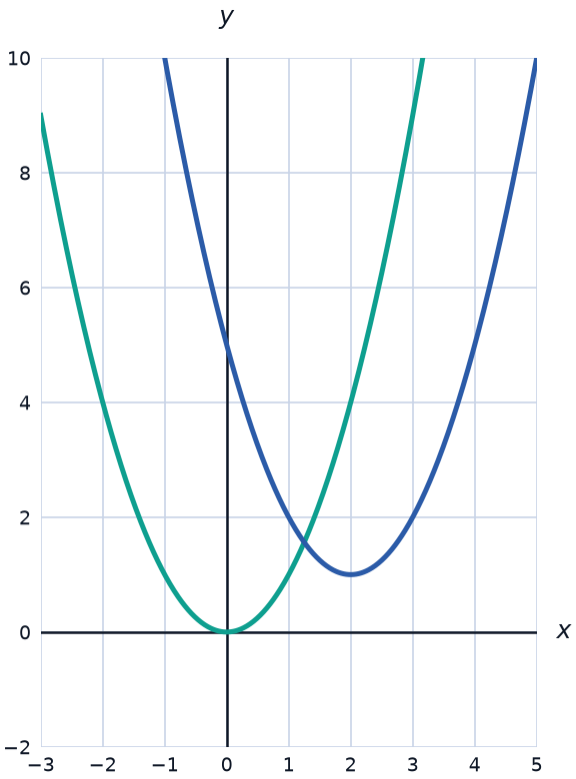
17 [Challenging] • Target: Exponential equations and logs as inverses • Cross-Lesson: 5.01 Trigonometry

The functions $f(x) = \log_3 x$ and $g(x) = 3^x$ are inverses. Use this fact to solve $3^x = 81$ without a calculator, and state the value of $g(f(16))$.

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18 [Challenging] • Target: Transformations of functions • Cross-Lesson: 6.01 Circles & Coordinate Geometry

The graph shows $f(x) = x^2$ (teal) and $g(x) = (x - 2)^2 + 1$ (indigo). Describe the transformation that maps f onto g.



19 [Challenging] • Target: Transformations of functions • Cross-Lesson: 7.01 Solid Geometry & Similarity

If $h(x) = -f(x + 3)$ where $f(x) = x^2$, describe the transformations applied to f to obtain h.

20 [Challenging] • Target: A well-designed word problem: investment growth and doubling time • Cross-Lesson: 8.01 Statistics & Probability

This question has three parts. An investment fund grows according to $A(t) = A_0 e^{0.045t}$, where A_0 is the initial amount (in dollars) and t is time in years. (a) If $A_0 = 8000$ dollars, find $A(10)$ to the nearest dollar. (b) Find the time t, to the nearest tenth of a year, for the investment to double. (c) Solve for t in terms of A and A_0 — that is, find the inverse function $t(A)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Domain and range

If $f(x) = -4x + 7$ and $g(x) = -2x - 5$, find $(f \circ g)(7)$.

a. 83

b. 85

c. -19

d. 81

22 [Foundational] • Target: Composite functions

If $f(x) = x$ and $g(x) = -3x + 3$, find $(f \circ g)(6)$.

a. -13

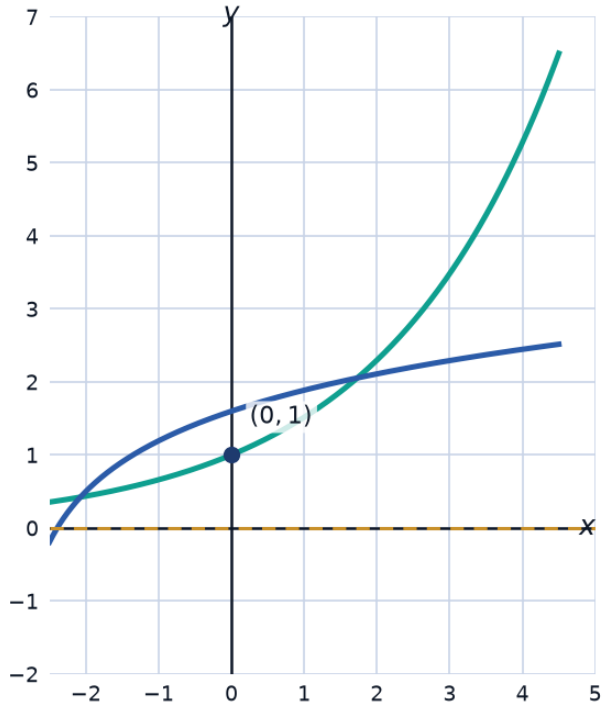
b. -15

c. -12

d. -17

23 [Foundational] • Target: Inverse functions

If $f(x) = 2x - 6$ and $g(x) = -3x + 1$, find $(f \circ g)(3)$.



a. -20

b. -22

c. -8

d. -24

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24 [Foundational] • Target: Verifying inverses

If $f(x) = x$ and $g(x) = 3x - 9$, find $(f \circ g)(4)$.

a. 6

b. 5

c. 1

d. 3

25 [Foundational] • Target: Logarithm laws

If $f(x) = x - 1$ and $g(x) = 2x + 4$, find $(f \circ g)(-4)$.

a. -3

b. -7

c. -4

d. -5

26 [Foundational] • Target: Solving logarithmic equations

If $f(x) = 4x + 8$ and $g(x) = 2x - 2$, find $(f \circ g)(-7)$.

a. -56

b. -16

c. -58

d. -54

27 [Foundational] • Target: Exponential equations and logs as inverses

If $f(x) = -4x + 5$ and $g(x) = 2x + 3$, find $(f \circ g)(-2)$.

a. 11

b. 7

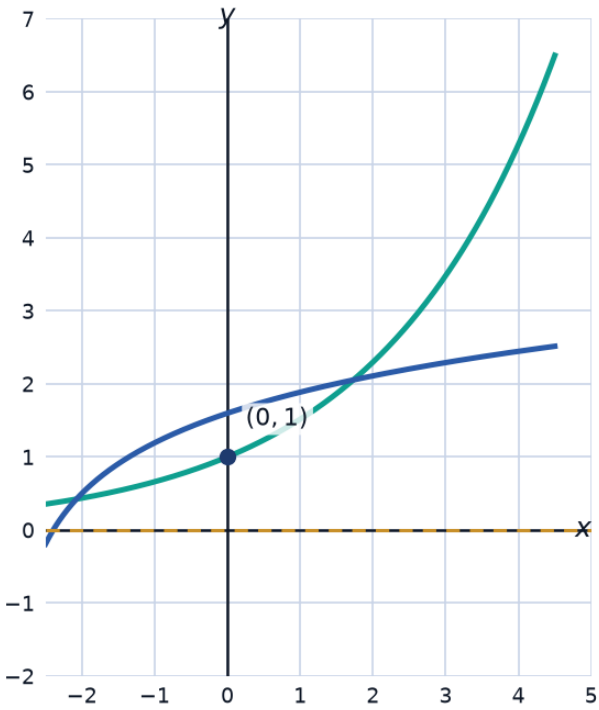
c. -1

d. 9

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28 [Foundational] • Target: Transformations of functions

Find $f^{-1}(x)$ for $f(x) = \frac{6x-5}{8}$.



a. $\frac{6x+5}{8}$

b. $\frac{8x+5}{6}$

c. $\frac{8x-5}{6}$

d. $\frac{6x-5}{8}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: investment growth and doubling time

Find $f^{-1}(x)$ for $f(x) = \frac{9x-13}{4}$.

a. $\frac{9x-13}{4}$

b. $\frac{4x-13}{9}$

c. $\frac{9x+13}{4}$

d. $\frac{4x+13}{9}$

30 [Intermediate] • Target: Inverses & Domain Restrictions

Find $f^{-1}(x)$ for $f(x) = \frac{5x-14}{3}$.

a. $\frac{5x-14}{3}$

b. $\frac{3x-14}{5}$

c. $\frac{3x+14}{5}$

d. $\frac{5x+14}{3}$

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31 [Intermediate] • Target: Composition & Decomposition of Functions

Find $f^{-1}(x)$ for $f(x) = \frac{7x - 15}{4}$.

a. $\frac{4x + 15}{7}$

b. $\frac{7x - 15}{4}$

c. $\frac{7x + 15}{4}$

d. $\frac{4x - 15}{7}$

32 [Intermediate] • Target: Logarithmic Laws & Base Changes

Find $f^{-1}(x)$ for $f(x) = \frac{5x - 2}{8}$.

a. $\frac{5x - 2}{8}$

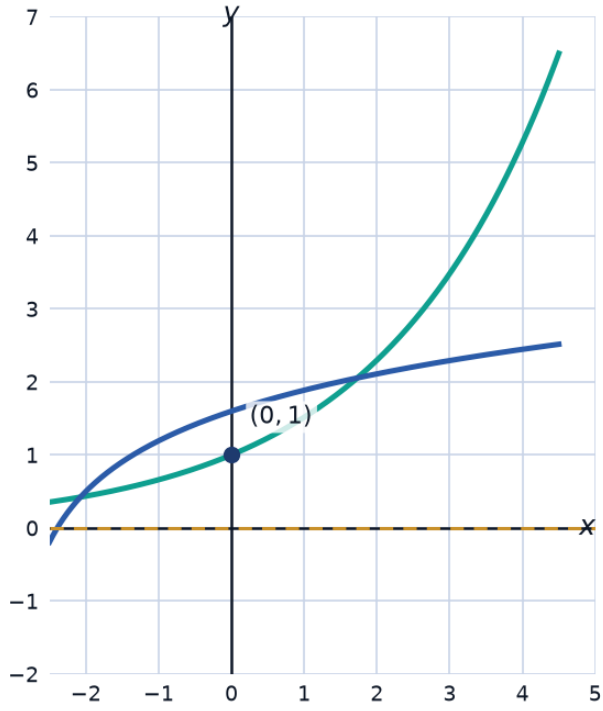
b. $\frac{8x - 2}{5}$

c. $\frac{5x + 2}{8}$

d. $\frac{8x + 2}{5}$

33 [Intermediate] • Target: Exponential Models & Logarithmic Graphs

Find $f^{-1}(x)$ for $f(x) = \frac{2x - 11}{6}$.



a. $\frac{2x - 11}{6}$

b. $\frac{2x + 11}{6}$

c. $\frac{6x - 11}{2}$

d. $\frac{6x + 11}{2}$

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34 [Intermediate] • Target: Domain and range

Find $f^{-1}(x)$ for $f(x) = \frac{6x-3}{3}$.

a. $\frac{6x-3}{3}$

b. $\frac{6x+3}{3}$

c. $\frac{3x-3}{6}$

d. $\frac{3x+3}{6}$

35 [Intermediate] • Target: Composite functions

Solve $\ln x + \ln(x-5) = \ln 6$.

a. 8

b. -1

c. 4

d. 6

36 [Intermediate] • Target: Inverse functions

Solve $\ln x + \ln(x-6) = \ln 247$.

a. -13

b. 17

c. 21

d. 19

37 [Intermediate] • Target: Verifying inverses

Solve $e^{3x-7} = 105$, giving your answer to three decimal places.

a. 3.385

b. 3.885

c. 4.885

d. 4.385

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Logarithm laws • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Solve $e^{4x-6} = 488$, giving your answer to three decimal places.

a. 4.048

b. 3.548

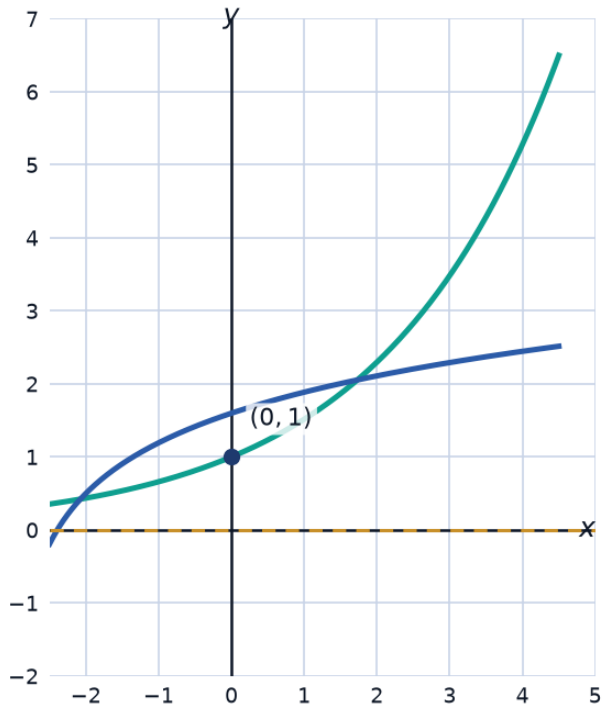
c. 3.048

d. 2.548

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39 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 7.01 Solid Geometry & Similarity

Evaluate $\log_5 500 - \log_5 4$ without a calculator.



a. 496

b. 3

c. 4

d. 2

40 [Challenging] • Target: Exponential equations and logs as inverses • Cross-Lesson: 8.01 Statistics & Probability

Evaluate $\log_3 72 - \log_3 8$ without a calculator.

a. 64

b. 2

c. 3

d. 1

41 [Challenging] • Target: Inverses & Domain Restrictions • Cross-Lesson: 1.01 Advanced Algebra

If $f(x) = \frac{2x-5}{3x+1}$, what is the value of $f^{-1}(1)$?

a. -6

b. -5

c. -4

d. -2

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42 [Challenging] • Target: Inverses & Domain Restrictions • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Given $f(x) = x^2 - 4x + 7$ defined on the restricted domain $[2, \infty)$, what is the equation of $f^{-1}(x)$?

a. $f^{-1}(x) = 2 + \sqrt{x - 3}$

b. $f^{-1}(x) = 2 - \sqrt{x - 3}$

c. $f^{-1}(x) = \sqrt{x - 7} + 2$

d. $f^{-1}(x) = 2 + \sqrt{x + 3}$

43 [Challenging] • Target: Composition & Decomposition of Functions • Cross-Lesson: 4.01 Lines, Triangles & Polygons

If $h(x) = (2x^2 - 1)^5$, identify inner function $g(x)$ and outer function $f(u)$ such that $h(x) = f(g(x))$.

a. $g(x) = 2x^2 - 1$ and $f(u) = u^5$

b. $g(x) = x^5$ and $f(u) = 2u^2 - 1$

c. $g(x) = 2x$ and $f(u) = (u - 1)^5$

d. $g(x) = 2x^2$ and $f(u) = u^5 - 1$

44 [Challenging] • Target: Logarithmic Laws & Base Changes • Cross-Lesson: 5.01 Trigonometry

Using the change of base formula, evaluate $\log_8(32)$.

a. $\frac{5}{3}$

b. $\frac{3}{5}$

c. 4

d. $\frac{4}{3}$

45 [Challenging] • Target: Exponential Models & Logarithmic Graphs • Cross-Lesson: 6.01 Circles & Coordinate Geometry

What is the vertical asymptote of the graph of $f(x) = 3\ln(2x - 8) + 1$?

a. $x = 4$

b. $x = 8$

c. $x = 0$

d. $y = 1$

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